

Homework Section 12.6

1.
 - a) Sketch one period of the curve given by $y = \sin x$ in the xy -plane.
 - b) Sketch one period of the cylinder given by $y = \sin x$ in R^3 .
 - c) Sketch one period of the cylinder given by $z = \sin x$ in R^3 .
2. Sketch the cylinder:
 - a) $z = e^y$
 - b) $\frac{x^2}{4} + \frac{y^2}{9} = 1$
3. Identify and sketch the graph of the quadric surface.
 - a) $\frac{x^2}{4} + \frac{y^2}{9} + \frac{z^2}{16} = 1$
 - b) $x = y^2 + \frac{z^2}{4}$
 - c) $z^2 = 1 + \frac{x^2}{4} + \frac{y^2}{9}$
 - d) $z^2 = \frac{x^2}{4} + \frac{y^2}{9}$
 - e) $z = \frac{y^2}{16} - \frac{x^2}{9}$
 - f) $y^2 + \frac{z^2}{9} - x^2 = 1$
4. Rewrite the following equation in a form that allows you to identify the quadric surface that it represents: $y^2 + z^2 + 4x + 4y - 10z + 29 = 0$.
5. As previously mentioned, multiplying an $n \times n$ matrix times an $n \times 1$ column vector produces an $n \times 1$ column vector. It is also true that multiplying an $m \times n$ matrix times an $n \times 1$ column vector produces an $m \times 1$ column vector. Find the order of the column vectors that result from the following products without actually doing the multiplication. AGAIN, DO NOT ACTUALLY PERFORM THE MULTIPLICATIONS YET.
 - a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} -4 \\ 10 \end{bmatrix}$
 - b) $\begin{bmatrix} -5 & 0 & 3 \\ 2 & -4 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$
6. Go ahead and find the above matrix products in parts (a) and (b) by hand.

Big Idea: Multiplying an $m \times n$ matrix A times an $n \times 1$ column vector can be thought of as a function that maps vectors in R^n to vectors in R^m . Loosely interpreted,

we think of A as a function whose input values are column vectors. This kind of function is called a **transformation**.

Example: The multiplication in 5(a) above maps a vector in R^2 to a vector in R^3 (i.e. it *transforms* the vector $\langle -4, 10 \rangle$ into the vector $\langle 16, 28, 40 \rangle$).

7. Fill in the blanks: The multiplication in 5(b) above maps a vector in _____ to a vector in _____, namely the vector _____ is *transformed* into the vector _____.